

DATA SCIENCE INTERNSHIP MONTH - 2

Exploratory Data Analysis on Customer Churn and Netflix Titles Datasets

Abstract

This internship project focuses on performing exploratory data analysis (EDA), data preprocessing, and feature engineering on two real-world datasets: the Telecom Customer Churn Dataset and the Netflix Titles Dataset. The primary goal was to understand dataset characteristics, clean inconsistencies, engineer meaningful features, and derive insights through statistical and visual analysis.

The first task analyzes customer churn in the telecom industry, identifying factors affecting customer retention. The second task explores Netflix content metadata to uncover trends in media production, content types, and country-wise contributions. Both tasks strengthened practical knowledge in Python, Pandas, NumPy, Matplotlib, and Seaborn.

Task 3: Customer Churn Data Preparation & EDA

1. Introduction

Customer churn refers to customers discontinuing a company's services. In telecom businesses, understanding churn patterns is crucial for improving retention and revenue. Data analysis helps identify the key reasons customers leave and supports future predictive modeling.

This task involved preparing telecom customer data through preprocessing and exploratory analysis to reveal meaningful business insights.

2. Objective

The objectives of this task were:

- To load and understand telecom customer churn data
- To clean missing and inconsistent values
- To engineer useful features for analysis
- To visualize customer behavior patterns
- To identify important factors influencing churn

3. Dataset Description

The telecom customer churn dataset contains customer account information, demographic details, subscribed services, billing information, and churn status.

First 5 records:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	\
0	No phone service	DSL	No	...	No	
1	No	DSL	Yes	...	Yes	
2	No	DSL	Yes	...	No	
3	No phone service	DSL	Yes	...	Yes	
4	No	Fiber optic	No	...	No	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
0	No	No	No	Month-to-month	Yes	
1	No	No	No	One year	No	
2	No	No	No	Month-to-month	Yes	
3	Yes	No	No	One year	No	
4	No	No	No	Month-to-month	Yes	

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

Dataset Characteristics

- Number of rows: approximately 7043
- Number of columns: 21
- Target variable: Churn
- Source: Kaggle

Important Features

- gender
- SeniorCitizen
- tenure

- MonthlyCharges
- TotalCharges
- Contract
- InternetService
- PaymentMethod
- Churn

4. Tools and Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- KaggleHub

5. Methodology

5.1 Data Collection

The dataset was loaded directly from Kaggle using KaggleHub.

5.2 Data Understanding

Initial exploration was done using:

- head()
- info()
- describe()
- isnull().sum()

This helped understand:

- data types

- missing values
- summary statistics

(7043, 21)

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	customerID	7043 non-null	object
1	gender	7043 non-null	object
2	SeniorCitizen	7043 non-null	int64
3	Partner	7043 non-null	object
4	Dependents	7043 non-null	object
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	object
7	MultipleLines	7043 non-null	object
8	InternetService	7043 non-null	object
9	OnlineSecurity	7043 non-null	object
10	OnlineBackup	7043 non-null	object
11	DeviceProtection	7043 non-null	object
12	TechSupport	7043 non-null	object
13	StreamingTV	7043 non-null	object
14	StreamingMovies	7043 non-null	object
15	Contract	7043 non-null	object
16	PaperlessBilling	7043 non-null	object
17	PaymentMethod	7043 non-null	object
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	object
20	Churn	7043 non-null	object

dtypes: float64(1), int64(2), object(18)

memory usage: 1.1+ MB

None

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure \
count	7043	7043	7043.000000	7043	7043	7043.000000
unique	7043	2	NaN	2	2	NaN
top	3186-AJIEK	Male	NaN	No	No	NaN
freq	1	3555	NaN	3641	4933	NaN
mean	NaN	NaN	0.162147	NaN	NaN	32.371149
std	NaN	NaN	0.368612	NaN	NaN	24.559481
min	NaN	NaN	0.000000	NaN	NaN	0.000000
25%	NaN	NaN	0.000000	NaN	NaN	9.000000
50%	NaN	NaN	0.000000	NaN	NaN	29.000000
75%	NaN	NaN	0.000000	NaN	NaN	55.000000
max	NaN	NaN	1.000000	NaN	NaN	72.000000

	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	\
count	7043	7043	7043	7043	...	
unique	2	3	3	3	...	
top	Yes	No	Fiber optic	No	...	
freq	6361	3390	3096	3498	...	
mean	NaN	NaN	NaN	NaN	...	
std	NaN	NaN	NaN	NaN	...	
min	NaN	NaN	NaN	NaN	...	
25%	NaN	NaN	NaN	NaN	...	
50%	NaN	NaN	NaN	NaN	...	
75%	NaN	NaN	NaN	NaN	...	
max	NaN	NaN	NaN	NaN	...	

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	\
count	7043	7043	7043	7043	
unique	3	3	3	3	
top	No	No	No	No	
freq	3095	3473	2810	2785	
mean	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	

	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	\
count	7043	7043	7043	7043.000000	
unique	3	2	4	NaN	
top	Month-to-month	Yes	Electronic check	NaN	
freq	3875	4171	2365	NaN	
mean	NaN	NaN	NaN	64.761692	
std	NaN	NaN	NaN	30.090047	
min	NaN	NaN	NaN	18.250000	
25%	NaN	NaN	NaN	35.500000	
50%	NaN	NaN	NaN	70.350000	
75%	NaN	NaN	NaN	89.850000	
max	NaN	NaN	NaN	118.750000	

	TotalCharges	Churn
count	7043	7043
unique	6531	2
top		No
freq	11	5174
mean	NaN	NaN
std	NaN	NaN
min	NaN	NaN
25%	NaN	NaN
50%	NaN	NaN
75%	NaN	NaN
max	NaN	NaN

[11 rows x 21 columns]

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

5.3 Data Cleaning

The following preprocessing operations were performed:

- Blank values in TotalCharges replaced with NaN
- Converted TotalCharges to numeric type
- Missing values filled using median
- customerID removed as irrelevant identifier

5.4 Feature Engineering

Two new features were created:

TenureGroup

Customer tenure categorized into ranges:

- 0–12 months
- 13–24 months
- 25–36 months
- 37–48 months
- 49–60 months
- 61–72 months

Step 5: Feature Engineering

1.Tenure Group

```
bins = [0,12,24,36,48,60,72]
labels = ['0-12', '13-24', '25-36', '37-48', '49-60', '61-72']

df['TenureGroup'] = pd.cut(df['tenure'], bins=bins, labels=labels)
```

2.Average Monthly Spend

```
df['AvgMonthlySpend'] = df['TotalCharges'] / (df['tenure'] + 1)
```

3.Binary Conversion

```
binary_cols = ['Partner', 'Dependents', 'PhoneService', 'PaperlessBilling', 'Churn']

for col in binary_cols:
    df[col] = df[col].map({'Yes':1, 'No':0})
```

Binary categorical values (Yes/No) were converted into 1/0 for analysis.

6. Exploratory Data Analysis

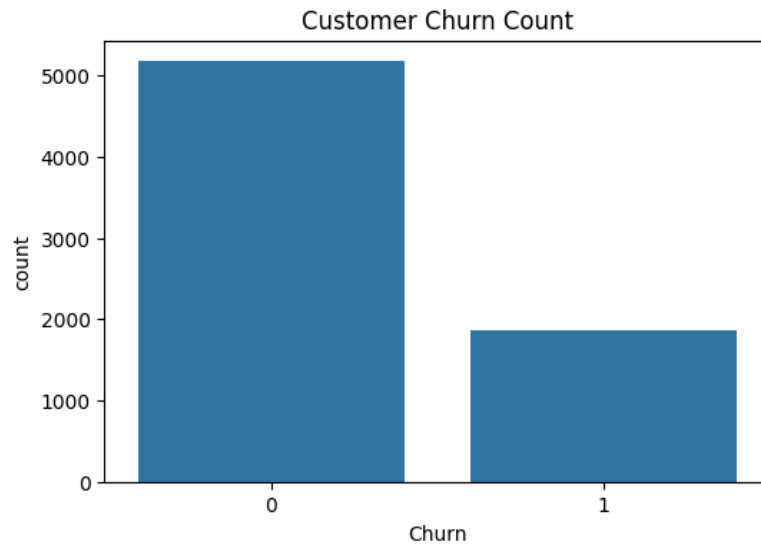
Several visualizations were generated to understand patterns.

6.1 Churn Distribution

A countplot showed the proportion of churned and retained customers.

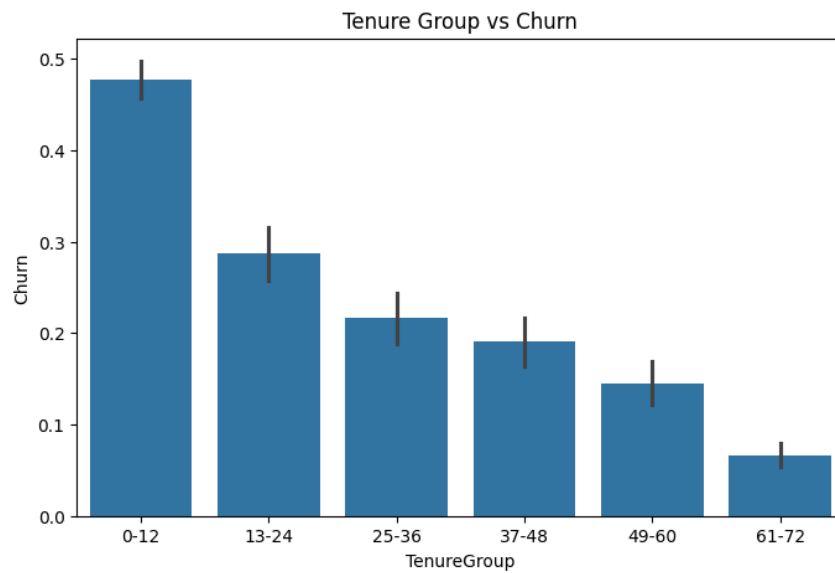
Observation:

- Majority customers remained active
- Significant percentage had churned



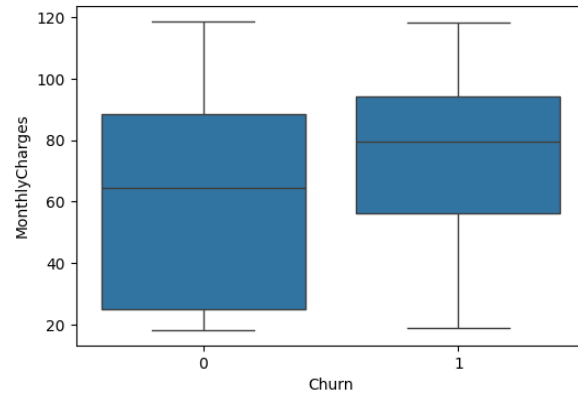
6.2 Tenure Group vs Churn

Customers with lower tenure showed higher churn rates.



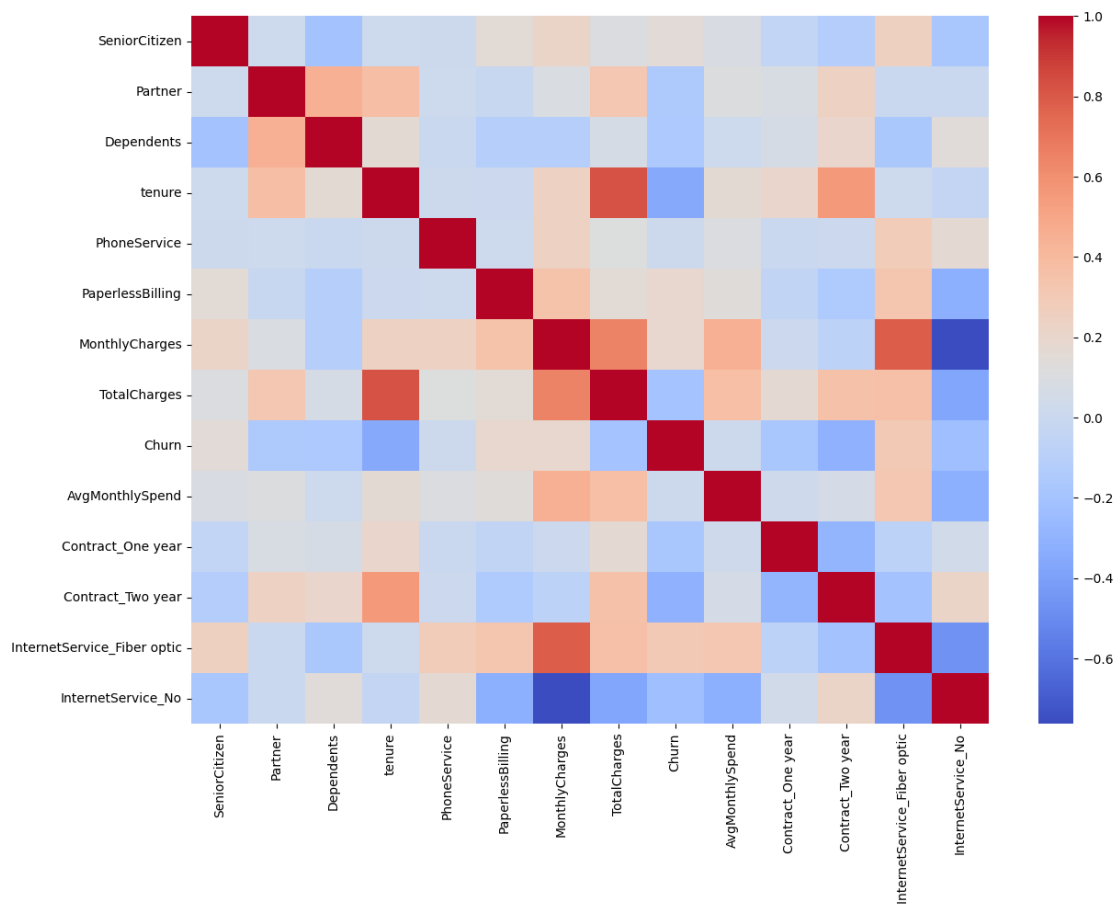
6.3 Monthly Charges by Churn

Boxplots indicated that customers paying higher monthly charges were more likely to leave.



6.4 Correlation Heatmap

Correlation analysis highlighted relationships



7. Results

The following insights were observed:

1. Customers with month-to-month contracts had higher churn.

2. High monthly charges were associated with churn.
3. Short-term customers were less loyal.
4. Long-term customers showed better retention.
5. Contract type strongly influenced customer decisions.

8. Conclusion

The telecom churn dataset was successfully cleaned and transformed for further machine learning applications. EDA revealed several business-relevant patterns that can help in customer retention strategies.

Task 4: Netflix Titles Dataset – Exploratory Analysis & Insights

1. Introduction

Streaming platforms like Netflix produce vast amounts of content. Analyzing content metadata helps understand trends in media production, popular genres, and regional contributions.

This task focused on performing EDA on the Netflix titles dataset.

2. Objective

The objectives were:

- To analyze Netflix content metadata
- To clean missing values
- To extract temporal and categorical insights
- To visualize global content trends

3. Dataset Description

The Netflix titles dataset contains metadata about movies and TV shows available on Netflix.

Dataset Characteristics

- Number of rows: approximately 8800
- Number of columns: 12
- Source: Kaggle

Main Features

- title
- type
- director
- cast
- country
- date_added

- release_year
- listed_in

4. Methodology

4.1 Data Collection

Loaded using KaggleHub from Kaggle.

```
(8807, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
None
```

	show_id	type	title	director	cast
count	8807	8807	8807	6173	7982
unique	8807	2	8807	4528	7692
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough
freq	1	6131	1	19	19
mean	NaN	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN

	country	date_added	release_year	rating	duration
count	7976	8797	8807.000000	8803	8804
unique	748	1767	NaN	17	220
top	United States	January 1, 2020	NaN	TV-MA	1 Season
freq	2818	109	NaN	3207	1793
mean	NaN	NaN	2014.180198	NaN	NaN
std	NaN	NaN	8.819312	NaN	NaN
min	NaN	NaN	1925.000000	NaN	NaN
25%	NaN	NaN	2013.000000	NaN	NaN
50%	NaN	NaN	2017.000000	NaN	NaN
75%	NaN	NaN	2019.000000	NaN	NaN
max	NaN	NaN	2021.000000	NaN	NaN

4.2 Data Cleaning

The following operations were performed:

- Converted date_added to datetime
- Filled missing values in director, cast, country
- Standardized categorical values

4.3 Feature Engineering

New columns created:

- release_decade
- is_movie
- year_added
- content_type

4.4 Country Popularity

The country column contained multiple labels. These were split and analyzed using value counts.

Step 6: Top 5 Countries

```
top_countries = df['country'].str.split(', ').explode().value_counts().head(5)
print(top_countries)
```

```
country
United States    3689
India            1046
Unknown          831
United Kingdom   804
Canada           445
Name: count, dtype: int64
```

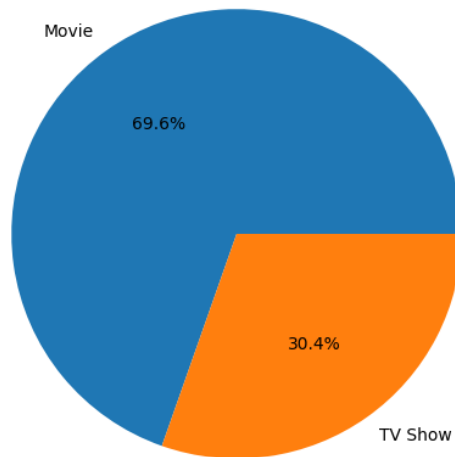
5. Exploratory Data Analysis

5.1 Movie vs TV Show Distribution

Pie chart revealed:

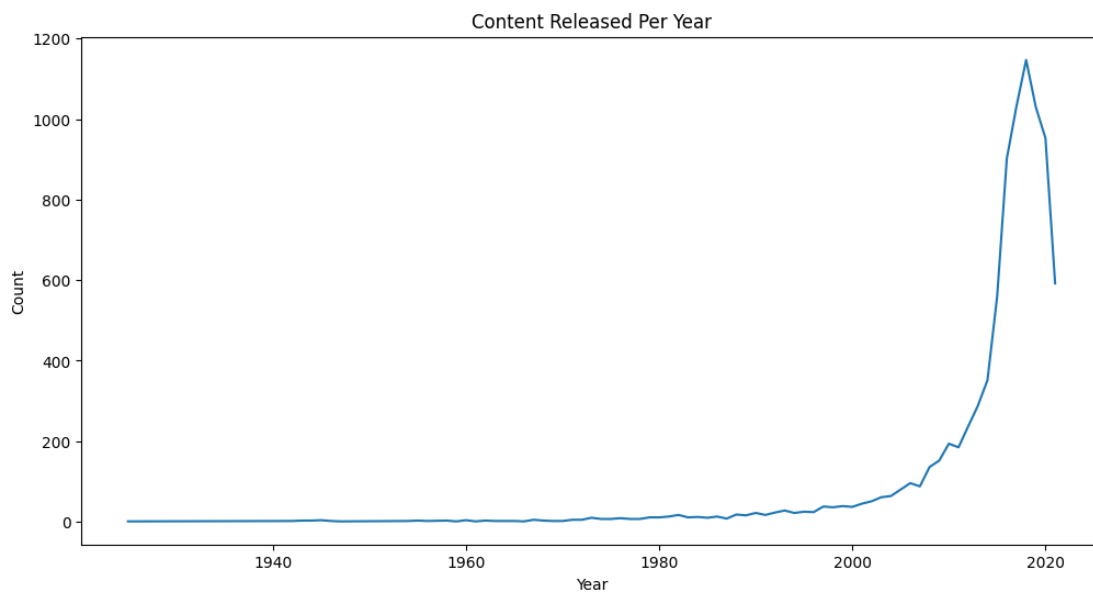
- Movies dominate catalog
- TV shows are smaller proportion

Movies vs TV Shows



5.2 Content Released Per Year

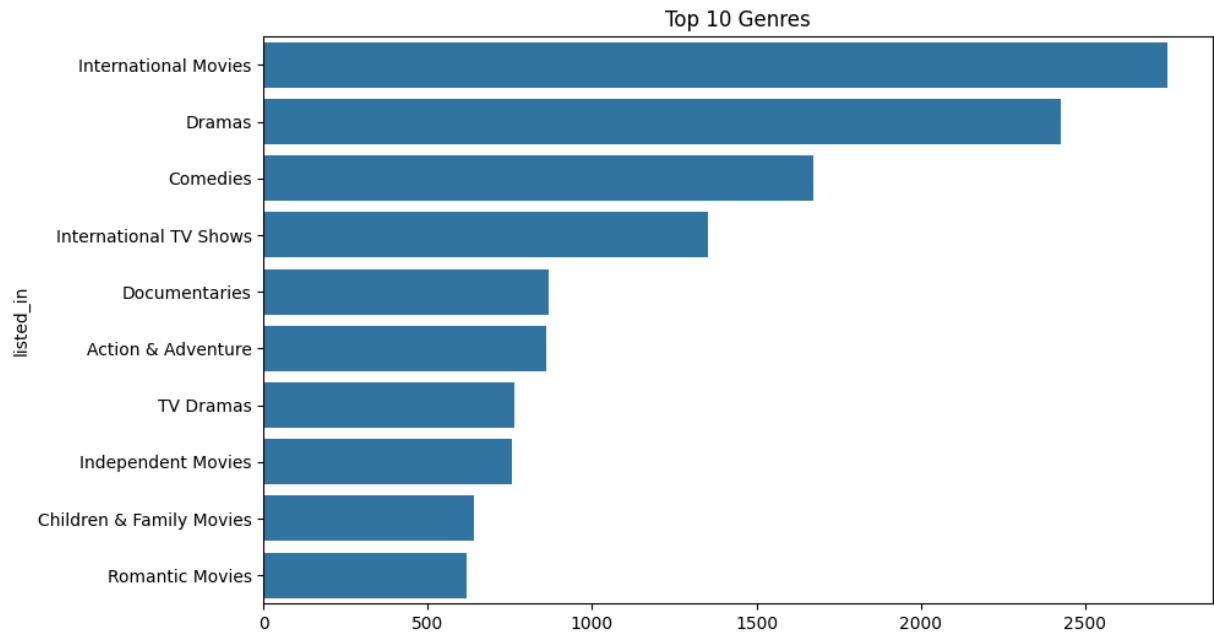
Line plot showed strong growth after 2015.



5.3 Top Genres

Most common genres included:

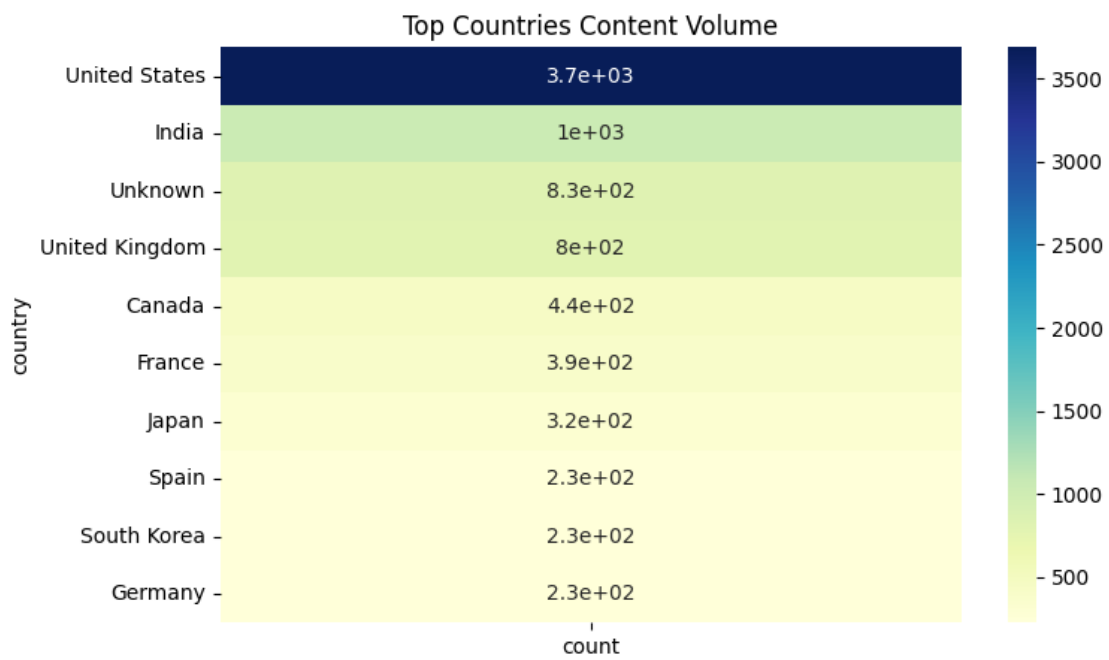
- International Movies
- Dramas
- Comedies



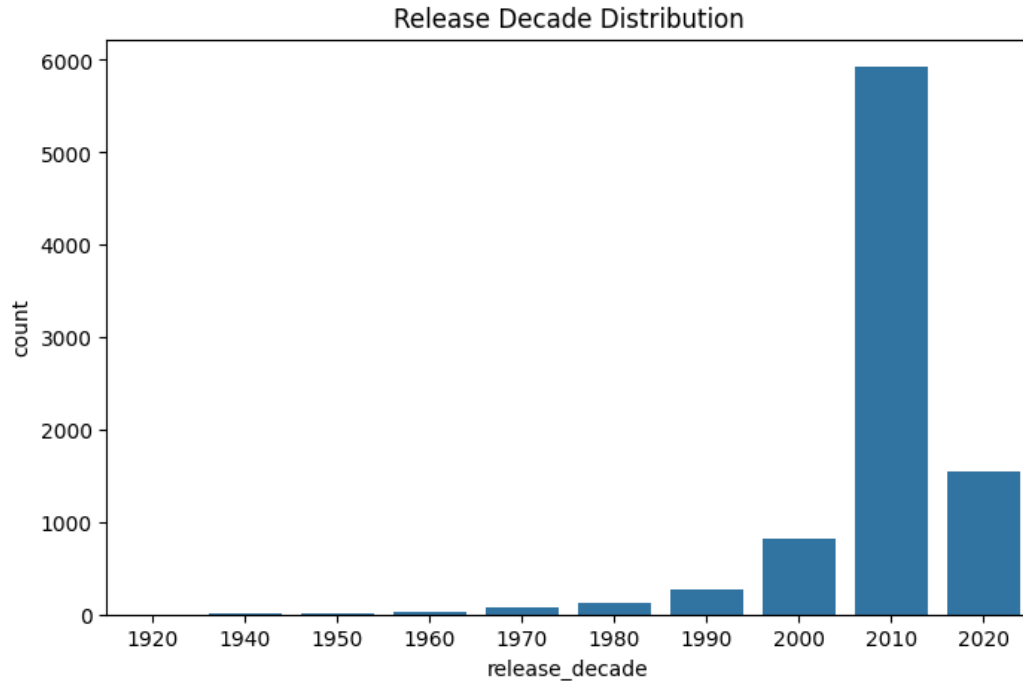
5.4 Country-Wise Analysis

Top contributing countries:

- United States
- India
- United Kingdom



6. Results



Key findings:

1. Movies are more common than TV shows.
2. Content production rapidly increased after 2015.
3. US contributes the most content.
4. India is among top-producing countries.
5. Drama and international categories dominate.

7. Trend Prediction

A simple linear regression model was applied to forecast content release for the next two years.

The model predicted continued growth in content availability.

8. Conclusion

The Netflix titles analysis successfully identified trends in media production and global content distribution. The project demonstrated how EDA supports decision-making in entertainment analytics.

Comparative Analysis of Both Tasks

Feature	Task 3	Task 4
Domain	Business	Entertainment
Dataset Type	Customer records	Media metadata
Main Focus	Churn analysis	Trend analysis
Techniques	Cleaning + EDA	Cleaning + EDA
Output	Retention insights	Content insights

Skills Learned

During these tasks, the following skills were developed:

- Data preprocessing
- Handling missing values
- Feature engineering
- Exploratory data analysis
- Statistical interpretation
- Visualization using Seaborn
- Working with real-world datasets

Challenges Faced

1. Missing values in columns
2. Incorrect data types
3. Multi-label categorical fields
4. Visualization interpretation
5. Dataset transformation

Overall Conclusion

The internship tasks provided practical exposure to data science workflows. Both projects emphasized the importance of preprocessing, feature extraction, and exploratory analysis for generating actionable insights from raw datasets. These tasks improved understanding of real-world analytics in both business and entertainment domains.